High Engine Temperature (HET) Shutdown Lamp Driver Malfunction

Symptoms

- The high engine temperature (HET) shutdown lamp is illuminated when a high engine temperature shutdown condition does **not** exist.
- The high engine temperature (HET) shutdown lamp is **not** illuminated when a pre-high engine temperature condition exists.

How To Use This Tree

This symptom tree can be used to troubleshoot a high temperature (HET) shutdown lamp driver malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check Customer Interface Box Wiring	
	STEP 1A. Check High Engine Temperature (HET) Shutdown Signal Wire for Open	Less than 10 ohms?
	STEP 1B. Check High Engine Temperature (HET) Shutdown Signal Wire for Wire to Wire Short	Less than 10 ohms?
	STEP 1C. Check High Engine Temperature Warning Signal Wire for Short to Ground	Less than 10 ohms?
STEP 2.	Check Engine Harness to Customer Interface Box Cable	

STEPS		SPECIFICATIONS
	STEP 2A. Check High Engine Temperature (HET) Shutdown Signal Wire for Open	Less than 10 ohms?
	STEP 2B. Check High Engine Temperature (HET) Shutdown Signal Wire for Wire to Wire Short	Less than 10 ohms?

STEP 1. Check Customer Interface Box Wiring

STEP 1A. Check High Engine Temperature (HET) Shutdown Signal Wire for Open

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box.

Action	Specification/Repair	Next Step
Check the high engine temperature (HET) shutdown signal wire for an open. • Place one test lead on the high engine temperature (HET) shutdown signal pin in connector C2. • Place the other test lead on high engine temperature (HET) shutdown signal terminal on the X4 connector.	Less than 10 ohms? **YES**	**1B**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html).	**Repair complete**

STEP 1B. Check High Engine Temperature (HET) Shutdown Signal Wire for Wire to Wire Short

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box.

Action	Specification/Repair	Next Step
Check the high engine temperature (HET) shutdown signal wire for wire to wire short. • Place one test lead on the high engine temperature (HET) shutdown signal pin in connector C2. • Place the other test lead on each of the remaining terminals in X4 connector.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html).	**Repair complete**
	Less than 10 ohms? **NO**	**1C**

STEP 1C. Check High Engine Temperature (HET) Shutdown Signal Wire for Short to Ground

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box.

Action	Specification/Repair	Next Step
Check the high engine temperature (HET) shutdown signal wire for short to ground. • Place one test lead on the high engine temperature (HET) shutdown signal pin in connector C2. • Place the other test lead on panel ground.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html).	**Repair complete**
	Less than 10 ohms? **NO**	**2A**

STEP 2. Check Engine Harness to Customer Interface Box Cable

STEP 2A. Check High Engine Temperature (HET) Shutdown Signal Wire for Open

Conditions :

- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box
- Disconnect customer interface box to engine harness cable connector C9 from the engine harness.

Action	Specification/Repair	Next Step
Check high engine temperature (HET) shutdown signal wire for an open. • Place a jumper between the pre-high engine temperature warning signal pin and the high engine temperature (HET) shutdown signal pin in the C9 connector. • Place one test lead in the pre-high engine temperature warning signal pin of the C2 connector. • Place the other test lead in the high engine temperature (HET) shutdown signal pin of the C2 connector.	Less than 10 ohms? **YES**	**2B**
	Less than 10 ohms? **NO** **Repair :** Replace the cable.	**Repair complete**

STEP 2B. Check High Engine Temperature (HET) Shutdown Signal Wire for Wire to Wire Short

Conditions :

- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box
- Disconnect customer inteface box to engine harness cable connector C9 from the engine harness.

Action	Specification/Repair	Next Step

Check high engine temperature (HET) shutdown signal wire for wire to wire short. • Place one test lead on the high engine temperature (HET) shutdown signal pin in connector C2. • Place the other test lead on each of the remaining pins in the C2 connector.	Less than 10 ohms? **YES** **Repair :** Refer to Section TF in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850 Modular Common Rail System Series Engines, Bulletin 4021493 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021493.html), or the Troubleshooting and Repair Manual, Electronic Control System, QSK38, QSK50, and QSK60, CM850 Modular Common Rail System Series Engines, Bulletin 4021533 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021533.html) or refer to the OEM service manual.	**Repair complete**
	Less than 10 ohms? **NO** **Repair :** Replace the cable.	**Repair complete**

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